

Solution Code - PL SQL programming - Cursors

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: solution.sql

```
1 | -- Scenario 1: GenerateMonthlyStatements
2 | DECLARE
3 |     CURSOR GenerateMonthlyStatements IS
4 |         SELECT t.TransactionID, t.AccountID, t.Amount, t.TransactionDate, c.Name
5 |         FROM Transactions t
6 |         JOIN Accounts a ON t.AccountID = a.AccountID
7 |         JOIN Customers c ON a.CustomerID = c.CustomerID
8 |         WHERE t.TransactionDate BETWEEN TRUNC(SYSDATE, 'MM') AND LAST_DAY(SYSDATE);
9 | BEGIN
10 |     FOR r IN GenerateMonthlyStatements LOOP
11 |         DBMS_OUTPUT.PUT_LINE('Customer: ' || r.Name || ' | Acc: ' || r.AccountID || ' | Trans: ' ||
r.TransactionID || ' | Amt: ' || r.Amount);
12 |     END LOOP;
13 | END;
14 | /
15 |
16 | -- Scenario 2: ApplyAnnualFee Cursor
17 | DECLARE
18 |     CURSOR ApplyAnnualFee IS SELECT AccountID, Balance FROM Accounts FOR UPDATE;
19 |     v_fee CONSTANT NUMBER := 20;
20 | BEGIN
21 |     FOR r IN ApplyAnnualFee LOOP
22 |         UPDATE Accounts SET Balance = Balance - v_fee WHERE CURRENT OF ApplyAnnualFee;
23 |     END LOOP;
24 |     COMMIT;
25 | END;
26 | /
27 |
28 | -- Scenario 3: UpdateLoanInterestRates Cursor
29 | DECLARE
30 |     CURSOR UpdateLoanInterestRates IS SELECT LoanID, InterestRate FROM Loans FOR UPDATE;
31 | BEGIN
32 |     FOR r IN UpdateLoanInterestRates LOOP
33 |         UPDATE Loans SET InterestRate = InterestRate + 0.5 WHERE CURRENT OF UpdateLoanInterestRates;
34 |     END LOOP;
35 |     COMMIT;
36 | END;
37 | /
```